* 0000800000001 *

Cambridge Assessment International Education

Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4941104148

MATHEMATICS (SYLLABUS D) **4024/11**

Paper 1 **October/November 2024**

2 hours

You must answer on the question paper.

You will need: Geometrical instruments

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.
- You may use tracing paper.
- You must show all necessary working clearly.

INFORMATION

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

© UCLES 2024

[Turn over

ELECTRONIC CALCULATORS MUST NOT BE USED IN THIS PAPER

1 At midday the temperature is −2 °C.
At 6 pm the temperature is 4 °C.

(a) Find the difference between these temperatures.

..... °C [1]

(b) The temperature at midnight is 9 °C lower than the temperature at midday.

Find the temperature at midnight.

..... °C [1]

2 Amber and Pablo share $280 in the ratio 2 : 5.

Work out Pablo's share.

$ ..... [2]

3 Here are eight integers.

−1 −5 −1 −3 −3 2 −1 −7

Find

(a) the mode

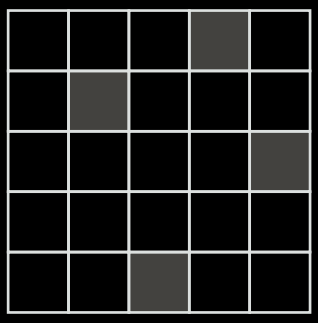
..... [1]

(b) the median.

..... [2]

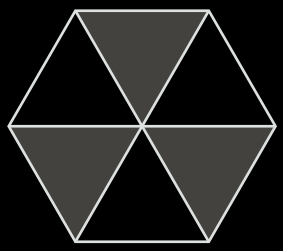
DO NOT WRITE IN THIS MARGIN

4 **(a)** Shade **one** more small square so the diagram has one line of symmetry.



[1]

(b) Here is a regular polygon.



Complete the description of the rotational symmetry of this polygon.

The polygon has rotational symmetry of order ..... [1]

5 **(a)** Simplify.

$2a - 3b + 4b - 5a$

..... [2]

(b) Expand.

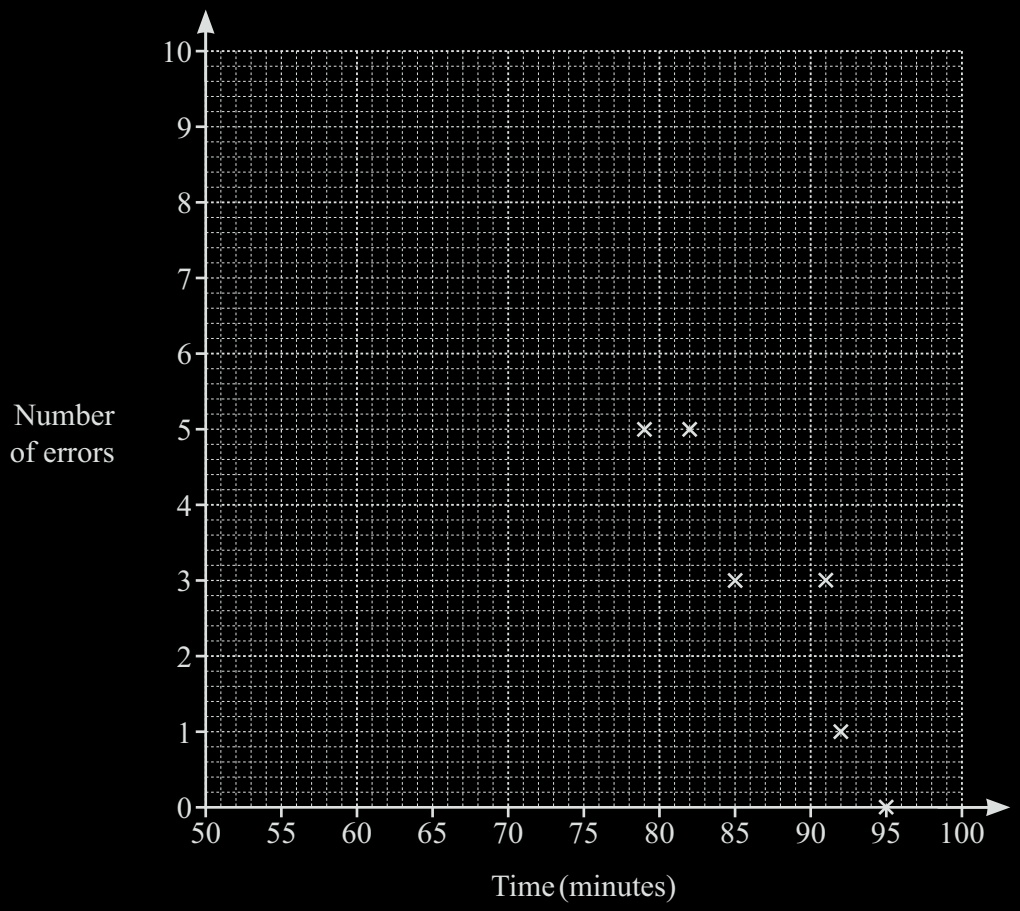
$5(3x - 2)$

..... [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

6 The table shows the time spent on a homework task and the number of errors made for some students in a class.

Time (minutes)	79	92	91	85	82	95	60	65	63	70
Number of errors	5	1	3	3	5	0	9	7	8	7



(a) Complete the scatter diagram.
The first 6 points have been plotted for you. [2]

(b) On the scatter diagram, draw a line of best fit. [1]

(c) Another of the students in the class made 6 errors.

Use your line of best fit to estimate the time this student spent on the homework task.

..... minutes [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

7 By writing each number correct to 1 significant figure, calculate an estimate for the value of

$$\frac{3.1 \times 26.7}{6.9 - 2.3}.$$

..... [2]

8 These are the first four terms of a sequence.

2 8 14 20

(a) Find the next number in the sequence.

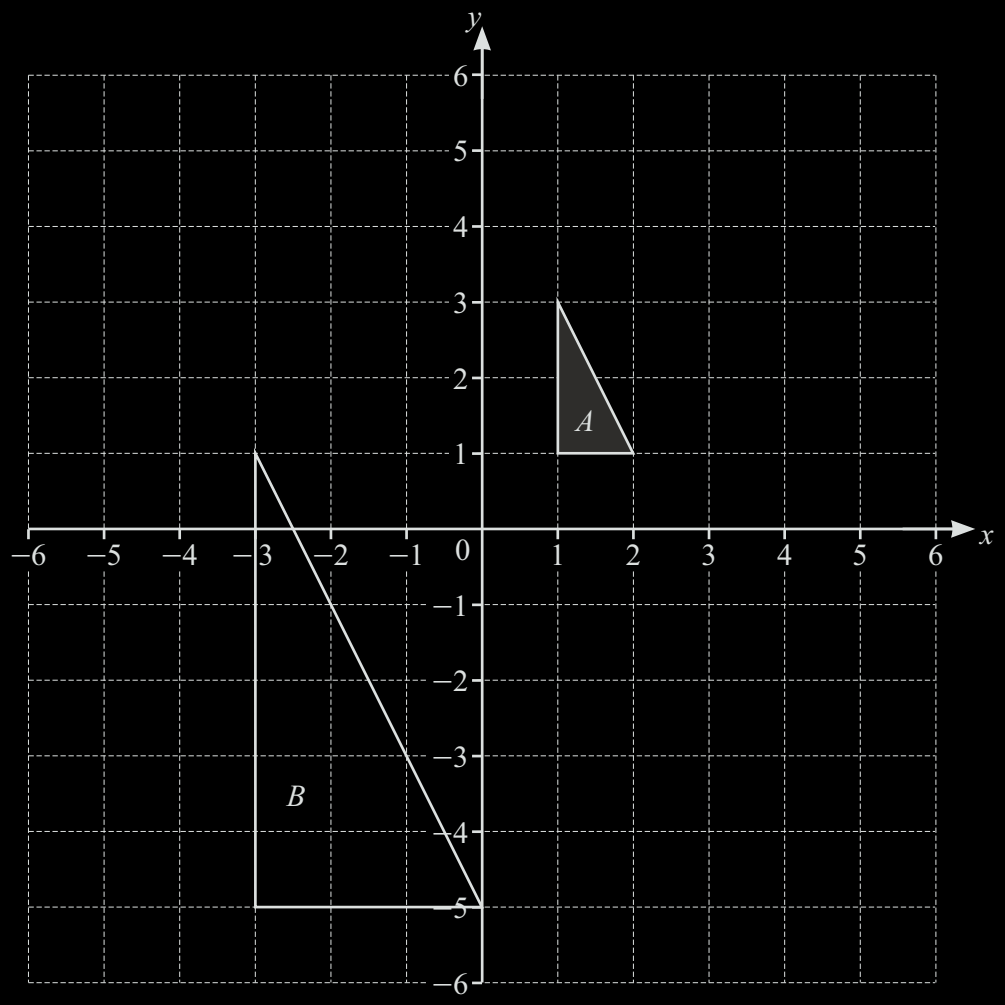
..... [1]

(b) Find an expression, in terms of n, for the nth term of this sequence.

..... [2]

DO NOT WRITE IN THIS MARGIN

9 The grid shows triangle A and triangle B.



(a) Describe fully the **single** transformation that maps triangle A onto triangle B.

.....

..... [3]

(b) Triangle A is mapped onto triangle C by the translation $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$.

Draw triangle C. [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

10 Solve the simultaneous equations.
Show your working.

$$3a + b = -4$$
$$2a + 3b = 9$$

$a =$ .....

$b =$ ..... [3]

11 Point A $(2, 4)$ is joined to point B $(5, -2)$ by a straight line.

(a) Find the coordinates of the midpoint of AB.

(..... , ...) [1]

(b) Find the gradient of the line AB.

..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

12 **(a)** Write 0.000 257 in standard form.

..... [1]

(b) Work out $\dfrac{2\times10^{4}}{4\times10^{-5}}$.

Give your answer in standard form.

..... [2]

13 Work out $2\frac{1}{5}\div\frac{3}{4}$.

Give your answer as a mixed number in its simplest form.

..... [2]

14 **(a)** Write 360 as a product of its prime factors.

..... [2]

(b) Find the smallest positive integer n such that $360n$ is a cube number.

$n =$ ..... [1]

DO NOT WRITE IN THIS MARGIN

15 A sector of a circle with angle 60° has arc length 4π cm.

Find the area of the sector.
Give your answer, as simply as possible, in terms of π.

..... cm^2 [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

16 **(a)** Express as a single matrix.

$$2\begin{pmatrix}3 & -1\\ 2 & 4\end{pmatrix}-\begin{pmatrix}1 & 3\\ -2 & 5\end{pmatrix}$$

$\begin{pmatrix} & \\ & \end{pmatrix}$ [2]

(b) $\mathbf{A}=\begin{pmatrix}4 & k\\ 2 & 1\end{pmatrix}$.

The determinant of **A** is 10.

(i) Find the value of k.

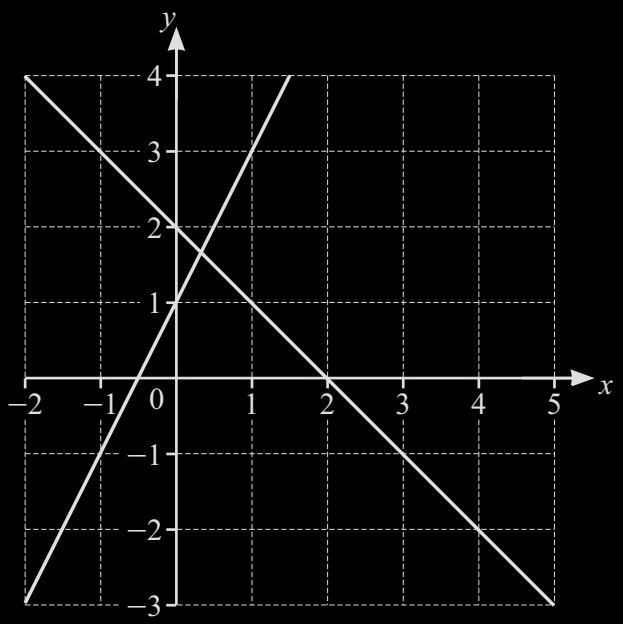
$k =$ ..... [1]

(ii) Find $\mathbf{A}^{-1}$.

$\mathbf{A}^{-1} = \begin{pmatrix} & \\ & \end{pmatrix}$ [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

17 The diagram shows the lines $y = 2x + 1$ and $x + y = 2$.



(a) The region R is defined by these three inequalities.

$$y \leqslant 2x + 1 \qquad x + y \leqslant 2 \qquad y \geqslant -2$$

On the diagram, shade and label the region R. [2]

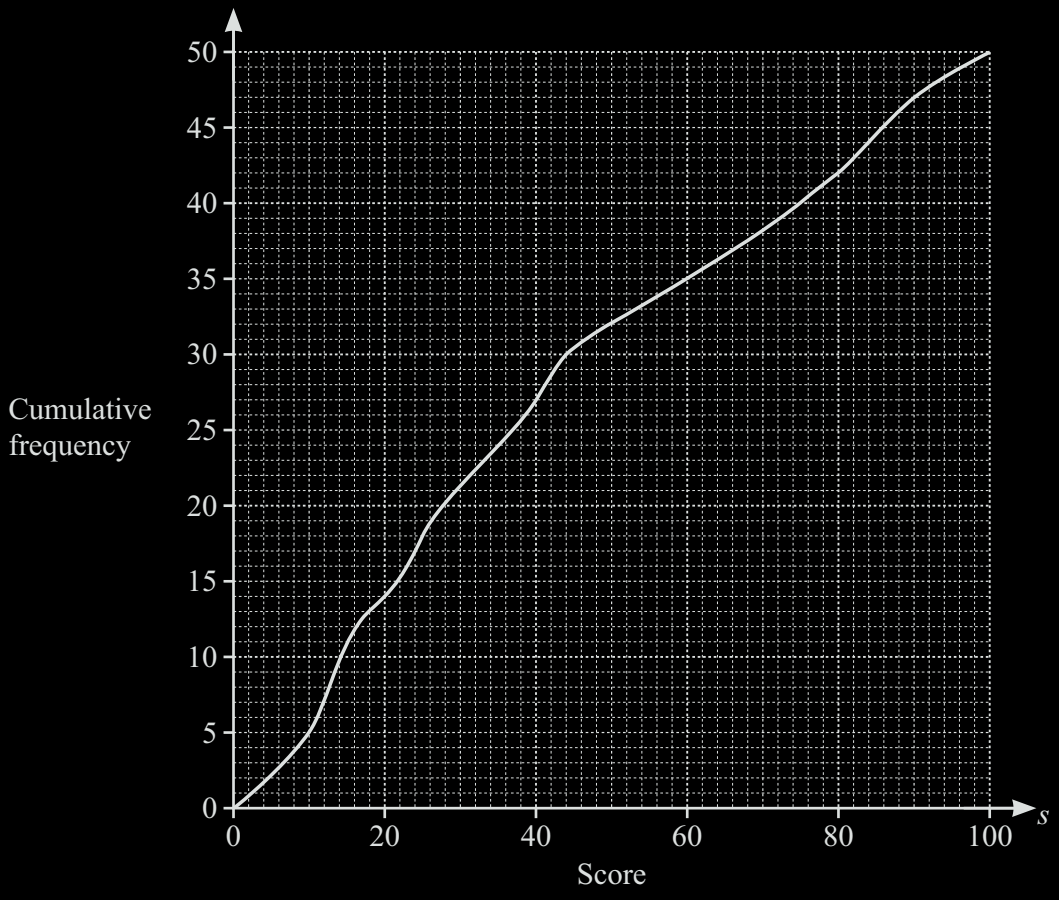
(b) The point $(k, k-2)$ lies in the region R where k is an integer.

List the possible values of k.

..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

18 50 adults each take part in a quiz.
The cumulative frequency diagram shows their scores.



(a) Use the diagram to find an estimate of the interquartile range.

..... [2]

(b) 20% of the adults win a prize for getting a high score in the quiz.

Use the diagram to work out the minimum score needed to win a prize.

..... [2]

(c) Use the diagram to complete the frequency table.

Score (s)	$0 < s \leqslant 20$	$20 < s \leqslant 40$	$40 < s \leqslant 60$	$60 < s \leqslant 80$	$80 < s \leqslant 100$
Frequency					8

[2]

DO NOT WRITE IN THIS MARGIN

19 x is inversely proportional to the square root of y.
When $x = 2$, $y = 16$.

Find y when $x = 32$.

$y =$ ..... [2]

20 $\left(ax^n\right)^{\frac{2}{3}} = 4x^{10}$

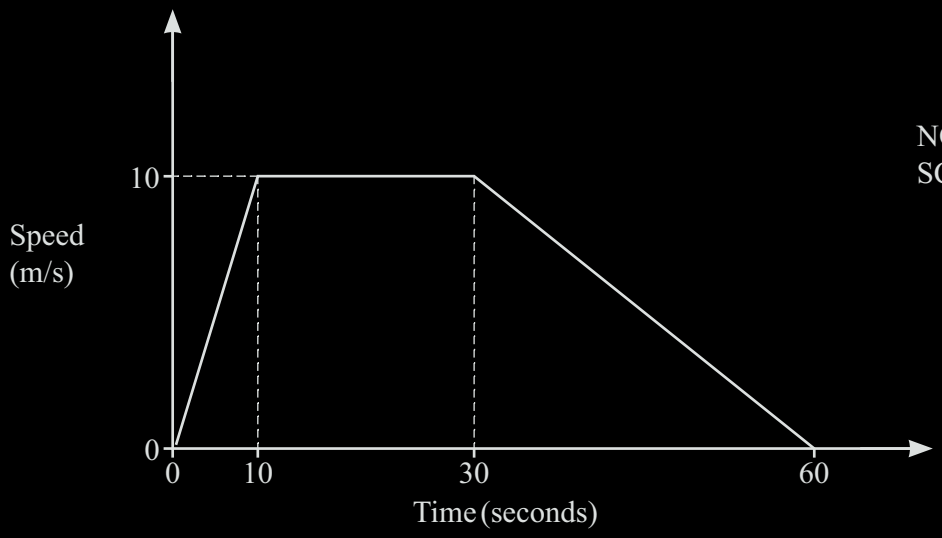
Work out the value of a and the value of n.

$a =$ .....

$n =$ ..... [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

21 The diagram shows the speed–time graph for a journey.



Calculate the total distance travelled.

..... m [2]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

22 $f(x) = 3x^2 + 5$

(a) Work out $f(-1)$.

..... [1]

(b) Solve $f(2x) = 17$.

$x =$ ..... or $x =$ ..... [3]

23 A rectangle has length 32 cm and width 15 cm.
Each measurement is given correct to the nearest centimetre.

(a) Write down the upper bound for the length.

..... cm [1]

(b) Calculate the upper bound for the difference between the length and the width.

..... cm [1]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

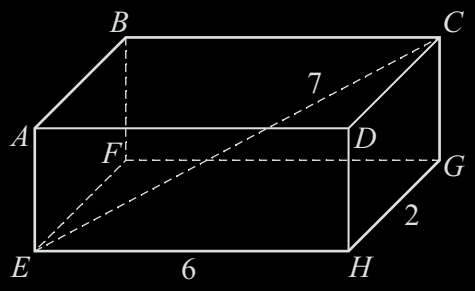
24 Simplify.

$$\frac{2x^2+5x-3}{2x^2+6x}$$

..... [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

25



NOT TO SCALE

The diagram shows a cuboid.
$EH = 6$ cm, $HG = 2$ cm and $EC = 7$ cm.

Calculate CG.

$CG =$ ..... cm [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

26 Solve.

$$\frac{x}{x-1} - \frac{3}{2x-1} = 1$$

$x =$ ..... [4]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

BLANK PAGE

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES 2024